EcoTubeHQ

EcoTubeHQ Video Player offers high quality video playback for many DRM-free streaming services whilst using minimal resources. This is achieved through minimal buffering, automated video codec selection, high quality video upscaling and hardware acceleration.

EcoTubeHQ offers AMD FidelityFX Super Resolution to produce high quality video output.

Unlike RTX VSR, AMD FSR works with most GPUs and iGPUs. Below is the official list of graphics cards that support AMD FSR:

AMD Ryzen desktop CPUs with AMD Radeon graphics	AMD Ryzen mobile CPUs with Radeon Graphics
AMD Radeon 6000 Series	AMD Radeon RX 500 Series
AMD Radeon 6000M Series	AMD Radeon RX 480, 470 and 460
AMD Radeon 5000 Series	Nvidia GeForce RTX 30 Series
AMD Radeon 5000M Series	Nvidia GeForce RTX 20 Series
AMD Radeon VII	Nvidia GeForce 16 Series
AMD Radeon RX Vega Series	Nvidia GeForce 10 Series

Minimal Buffering

EcoTubeHQ utilizes a 10 second minimal buffer to minimize data retrieved during video playback. In comparison, YouTube can retrieve up to 90 seconds of data for its buffer, and this can result in a lot of wasted data.

Here's how EcotubeHQ minimizes the amount of data buffered...

Let's say EcoTubeHQ outputs 31 seconds of a five minute video before playback is stopped. This means three 10 second buffers have been fully downloaded and output, and the fourth was fully downloaded and stopped after one second, thus leaving nine seconds of wasted data.

Figure 1: 9 seconds of wasted data [shown in grey]

If we stop playback of the same video after 31 seconds of conventional buffering there is approx 50 seconds of wasted data stored in the PC's buffer.

Figure 2: 50 seconds of wasted data [shown in grey - not to scale]

All of this data requires energy for it to be retrieved from YouTube's server and transported to your PC, so EcoTubeHQ's minimal buffer is saving time and energy by eliminating the need to transport an additional 41 seconds of wasted data.

This 41 seconds of data may seem like a tiny amount, but when you consider users watch over 1 billion hours of YouTube videos every day it is clear that the total amount of daily data wastage is mind boggling.

EcoTubeHQ's minimal buffer has the potential to significantly reduce data wastage.

Video resolution

The minimum bitrate for 720p resolution video is approx 3Mbps, whereas the minimum bitrate for 1080p video is approx 5Mbps. Let’s assume most people stream videos @1080p. Since the maximum resolution offered by EcoTubeHQ is 720p it uses approx 2Mbps less data for every second of video viewed.

To make up for the reduced quality of 720p video EcoTubeHQ uses AMD FidelityFX Super Resolution to perform upscaling, and this gives output which is roughly equivalent in quality to 1080p video which is using standard upscaling.

Frame Rate

Viewing 720p video @60fps typically requires 50%+ more bitrate than 30fps. Since the aim of EcoTubeHQ is minimizing data use it does not output video @60fps.

Supported Video Codecs

av1 - Best quality video with lowest bitrate

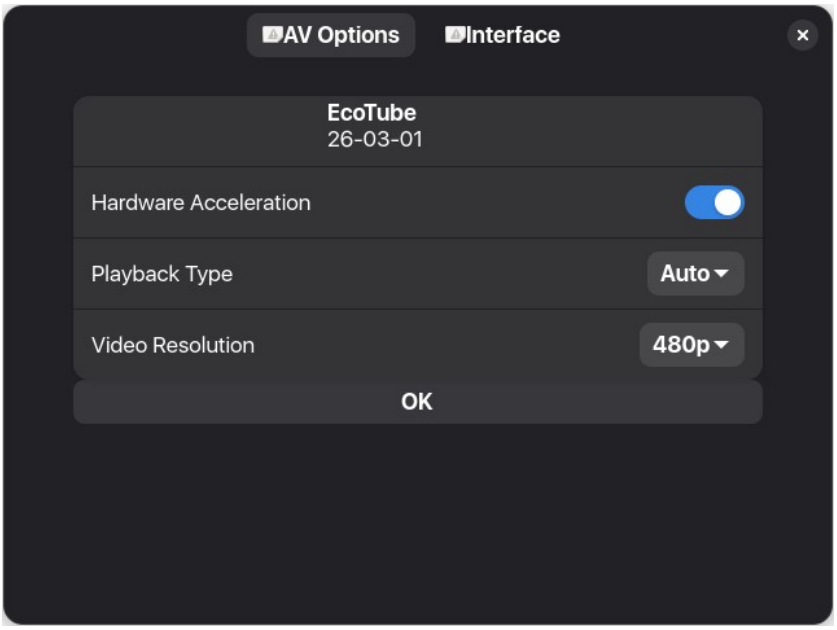
h.265 – Very good quality video with low bitrate

vp9 - Good quality video with low bitrate

h.264 - Lowest quality video with the highest bitrate

Player Preferences

Click on the hamburger menu next to the minimize button, then select Preferences. This shows the following…



...here's what it all means:

Hardware Acceleration

On [Default] - Vulkan GPU Compute Acceleration minimizes CPU use by providing hardware accelerated decode for av1, vp9, h.264 and h.265 codecs. This can improve playback on older PCs and improve the battery life of laptops. Turn off if playback issues occur.

Playback Type

Auto [Default] – This setting defaults to Quality on desktop computers as it is for use with laptops. If mains electricity is powering a laptop then the playback type defaults to **Quality**, whereas the playback type defaults to **Powersave** if battery power is being used.

Quality – av1 video codec is utilized with AMD FidelityFX Super Resolution upscaling to provide maximum video quality [if av1 is not available then vp9 is used, and if vp9 is not available then h.264 is used]. Debanding is also employed to show colors as smooth gradients rather than distinct bands, further improving picture quality [example image here].

Powersave vp9 video codec is utilized with Nearest Neighbor upscaling to reduce CPU usage [if vp9 is not available then h.264 is used].

Video Resolution

Supported Video Resolutions: 720p, **480p [Default],** 360p, 240p, 144p

When using the **Quality** playback type EcoTubeHQ restricts the maximum video resolution to 720p@30fps to minimize data usage. There is minimal difference when comparing EcoTubeHQ 720p **Quality** upscaling to 1080p browser output quality, but there is a significant reduction in data usage.

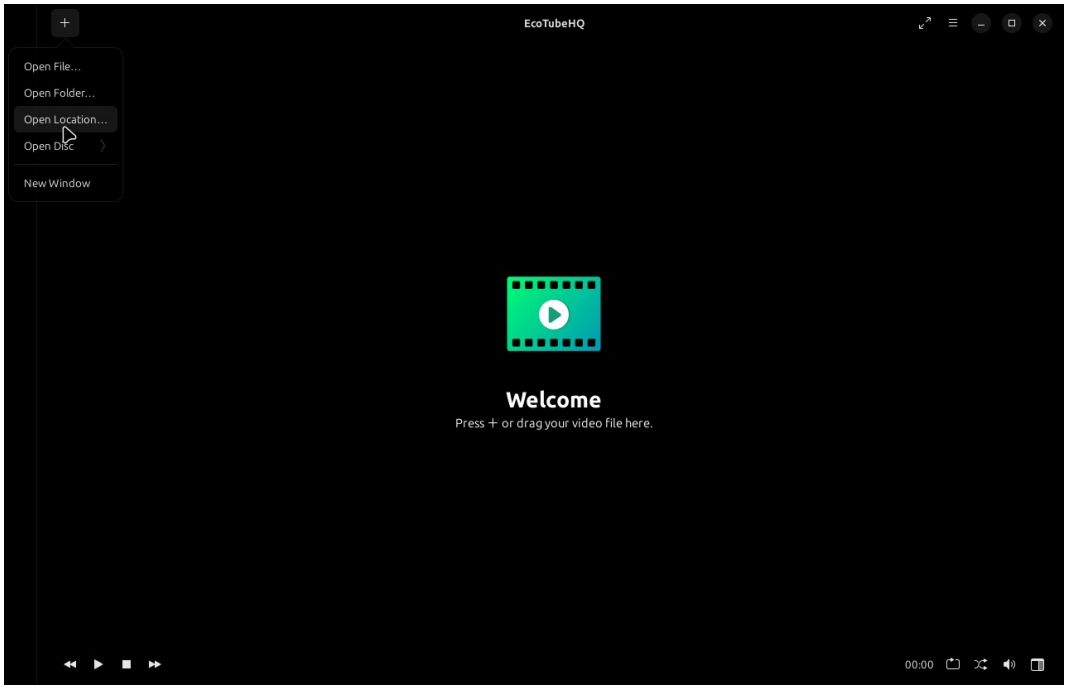
The maximum resolution of the **Powersave** playback type is 480p.

Note: Some h.264 videos on YouTube only have resolutions of 144p, 360p & 720p. If this is the case and the player's output resolution is set to 240p then the video output defaults to 360p, and if its output resolution is set to 480p then the video output defaults to 720p.

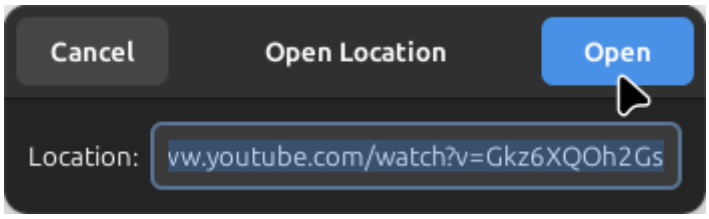
Note: ***EcoTubeHQ automatically updates yt-dlp and FFmpeg to minimize issues.***

How to use

To play a video from an online streaming service simply copy the video URL from the browser, click '+' at the top left of the player and select 'Open Location'…



...the player automatically places the copied video location into the 'Location' entry box…



...click the 'Open' button to initiate playback.

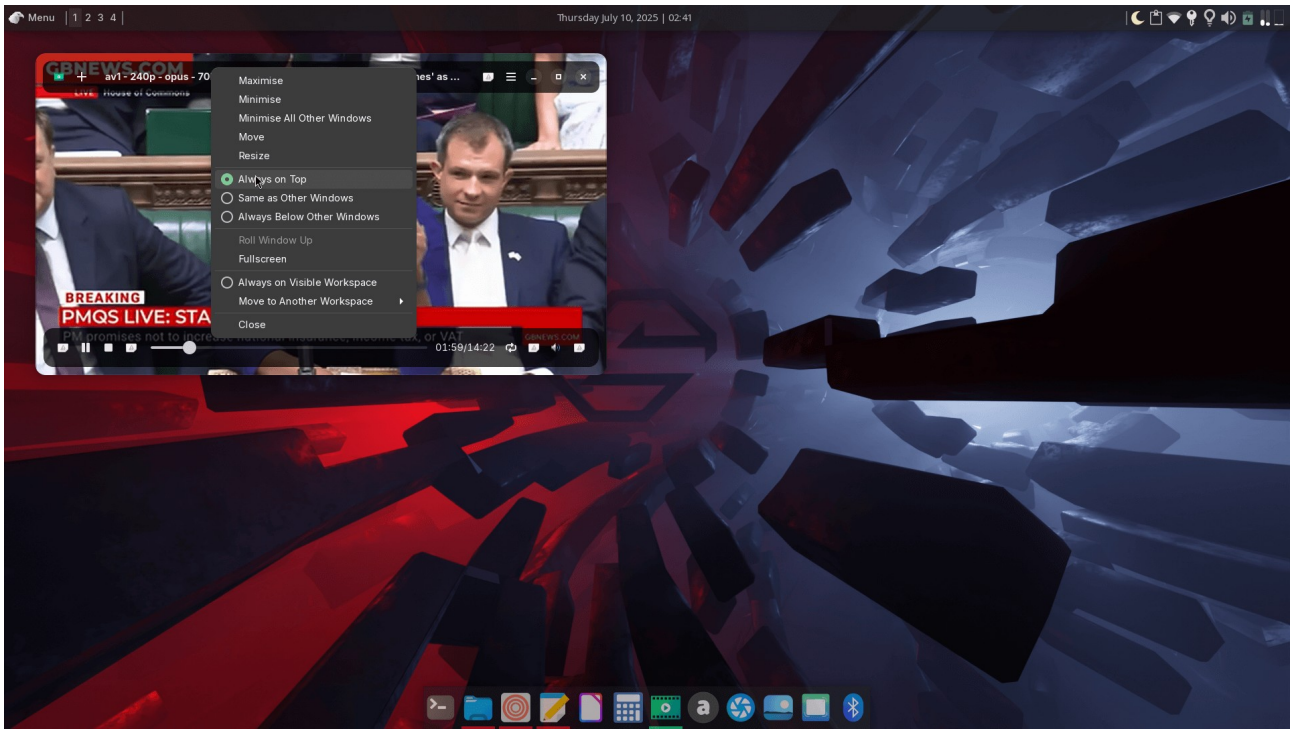
- Press the space bar or right click the mouse on the player screen to pause / play video playback.
- Rotate the mouse wheel on the playback window to turn the audio volume up / down – the volume will reset to normal when the player is restarted.
- Double click the player window during playback to play the video fullscreen.
- Press ‘Esc’ or double click on the player window during fullscreen playback to return the player to non-fullscreen playback.

Miniplayer Usage

If you are watching a YouTube video which does not require fullscreen or you want to minimize CPU usage during playback then set the player's video output resolution to 144p or 240p.

This automatically reduces the size of the player's non-fullscreen window, effectively providing you with a miniplayer. The size of 240p videos is shown in the screenshot below, and 144p videos are displayed in a smaller window.

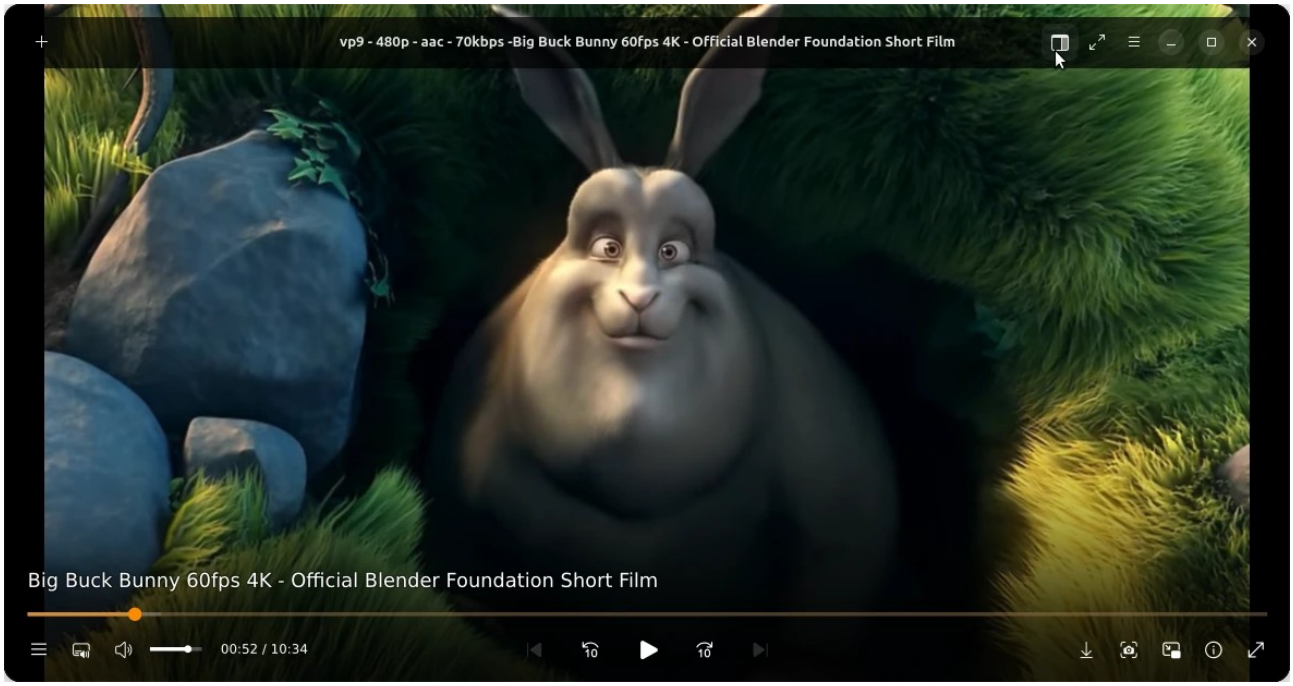
To keep the player on top of all other windows during playback right-click on the top bar and select 'Always on top' as shown below.



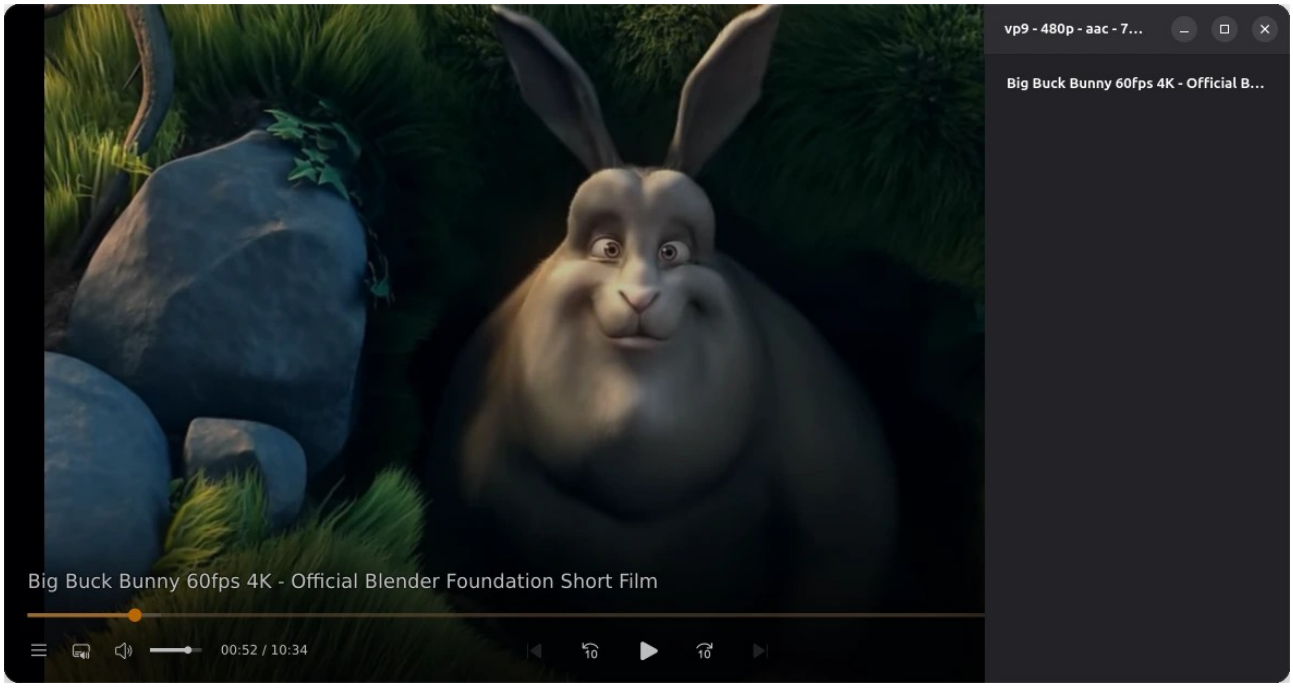
The end result minimizes CPU usage, and it significantly reduces the bandwidth of videos for things such as podcasts where video quality is not paramount.

* To further reduce CPU usage set the Playback Type to **Powersave**

Playlists



To add a video to a playlist click on the top icon indicated above. When this is done the playlist opens.



To add videos from an external streaming service drag and drop the browser URL for the playlist onto the player’s playlist. Additional videos can be added prior to, and during playback. To start playback simply double-click on the required playlist video.

Screenshots

To take a screenshot of the on-screen picture click the button at the bottom of the player shown below…

…the image will be stored in the **Pictures** folder.

Audio and video information

To see information relating to the on-screen video click the information button shown below…

EcoTubeHQ Original Concept: Colin Bett

Coding and additional ideas: Sako Adams

Forked from the Celluloid Video Player

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Changelog

v26.02.01

- Updated preferences and removed deprecated features
- Fixed dark theme issue
- Fixed issue occurring on Wayland on Open Location option
- Auto update ffmpeg

v25.12.02

- Add option to set many seconds of audio/video to prefetch
- Added mpv reload script
- Update ecotube fidelityfx super resolution logo

v25.12.01

- Fixed thumbnail on sandbox
- Disabled video download
- Enhance interface
- Reorganized Auto mode

v25.11.01

- Enhance interface
- Added modern OSC with screenshot and stats
- Fix FSR when plugged in
- Add ‘auto’ mode in desktop instance

v25.10.2

- yt-dlp is auto-updated when a new release is available
- Added modern On Screen Control
- Local files show thumbnails on the seekbar
- Download the video
- Enhance UI

v25.10.1

- Added ‘auto mode’ in playback type
- Fix stream latency
- Enhance UI

v25.9.3

- Updated yt-dlp
- Bilinear upscaling is used for powersave mode
- Enhance UI

v25.9.2

- Fix codec selection in powersave mode
- Updated playlist mode
- Enhance UI

v25.9.1

- Updated playback options - Powersave and Quality
- Updated playlist mode
- Added HDR option in preference
- Enhance UI
- Fixed memory usage in startup

v25.8.2

- Added two playback options - Powersave and Quality
- Removed redundant codec selection
- Reorganized default playlist
- Included no https protocol in format selection
- Reduced network stream usage

v25.8.1

- Forced yt-dlp to use MPEG-DASH on some resolutions
- Reduce stream when paused
- Fixed resolution dropping
- Exclude HDR
- Add playback type

v25.7.2

- Forced mpv to avoid HLS - temporarily

v25.7.1

- Added Vulkan support
- Fixed video / audio download + improved video loading
- Fix format selection
- Updated caching system
- Fixed auto-resize player

v25.5.1

- Upgrade to gtk4.16 and libadwaita-1 1.6
- Enhance interface with playlist overlay
- New update notification
- New dialog for "How to use it"
- Minor playback fix

v25.3.3

- Buffering: prefetch is now limited to 10 seconds to minimize data used when videos are not played in their entirety.
- Hardware acceleration is automatically activated if a laptop is running on battery power.
- Auto update parsers.
- Updated the way default resolutions are selected.
- The preference interface has been updated.

V25.2.1

26 February 2025

Initial conversion from Celluloid.

Thanks to haggen88 for the icon update and albanobattistella for the patches,

14 August 2024
Initial commit to Github.